

UNIVERSITY OF CLAUDE · COURSE SYLLABUS

CSCE 650 — Virtual Reality

Perception, tracking, latency, and the design of experience that does not make people ill

AT A GLANCE

Credits: 3 credit-equivalents · **Modules:** 13 · **Term:** Semester 4 (with CSCE 748 and CSCE 735)

Prerequisites: CSCE 641 Computer Graphics; CSCE 649 helpful for the interaction work; no hardware assumed beyond a consumer headset

Effort: 10–12 hours per week · 13 modules + 2 project weeks

Deliverable: A VR application that holds its frame budget, tracks correctly, renders stereo without error, and has been tested on people other than you — with their comfort ratings reported

Course description

Every other course in this program had a correctness criterion you could compute. A renderer converges to the rendering equation; a simulator conserves momentum; a database survives kill -9. **Virtual reality's criterion is a claim about a person** — whether they feel present, and whether they feel ill — and that changes the discipline.

The central fact is that **you are rendering to the vestibular system as much as to the eye**. A frame delivered 40 ms late is not a performance problem; it is a conflict between what the inner ear reports and what the eyes show, and the body resolves that conflict by inducing nausea. This is why VR has a hard real-time requirement that ordinary graphics does not, and why 'it runs at 45 fps, close enough' is not an acceptable sentence.

The second fact is that **most of the hard problems are perceptual rather than computational**. Why a corridor feels wrong, why a hand that passes through a table breaks presence, why teleport locomotion is comfortable and smooth locomotion is not, and why some people are fine and others are not — none of these are answered by profiling. They are answered by knowing how human perception works and by testing on humans.

You will build in a real engine rather than from scratch, because the compositor, the timewarp, and the driver stack are not things it is useful to reimplement. **The work is in the parts the engine does not do for you**, which is most of what determines whether an application is good.

Learning outcomes

On finishing this course you should be able to do the following without reference material.

- 1 Explain presence and the conditions that sustain and break it.
- 2 Apply the relevant properties of human vision to display and rendering decisions.
- 3 Implement correct stereo rendering with asymmetric frusta.
- 4 Explain lens distortion correction and why it interacts with timewarp.
- 5 Explain tracking, sensor fusion, and prediction.
- 6 Account for every millisecond of the motion-to-photon budget.
- 7 Explain cybersickness mechanistically and design to reduce it.
- 8 Compare locomotion schemes against the comfort/presence trade.
- 9 Apply VR-specific rendering optimisations and say what each costs.
- 10 Design and run an honest evaluation on real users.

Module schedule

One module per week for thirteen weeks, then two weeks for the final project. Each module is a slide deck, a set of lecture notes, mapped external lectures, and exercises that must run.

Wk	Module	Core question it answers
1	Presence, and What VR Must Deliver	What makes virtual reality work, and what makes it fail?
2	Human Vision for VR	What does the eye require, and what can you get away with?
3	Stereo Rendering	How do you render correctly for two eyes?
4	Display Optics and Distortion	Why does the rendered image look nothing like the submitted one?
5	Tracking	How does the system know where you are?
6	Latency	Where do the milliseconds go, and how do you get them back?
7	Cybersickness	What makes VR nauseating, and what can you do about it?
8	Locomotion	How do you let someone walk a kilometre in a 3 m room?
9	Interaction and Input	How do you let someone manipulate things that are not there?
10	Rendering for VR	How do you render two eyes within the budget?
11	Spatial Audio	Why does audio matter more in VR than anyone expects?
12	Augmented Reality and Passthrough	What changes when the real world is still visible?
13	Designing and Evaluating VR	How do you know whether what you built is any good?

Projects

Project 1 — A scene that holds its budget (after Module 7)

No interaction, no locomotion. A static scene the user can look around in, rendered correctly and comfortably. Everything later depends on this being right, and it is harder than it sounds.

Requirements

- Correct stereo with asymmetric frusta and the user's measured IPD, not a hard-coded default.
- Frame budget held at the headset's native rate, with per-eye GPU time reported and headroom stated.
- Measured motion-to-photon latency, by whatever method you can manage, with the method described.
- No frame drops over a five-minute session. Report the frame time distribution, not the mean.
- A deliberate comparison: the same scene with a correct IPD and with one 15 mm wrong, described from the inside.

Done means

- **A frame time histogram over five minutes with the 99th percentile under the budget.** Means hide exactly the frames that cause discomfort.
- Six people report no discomfort after five minutes, on a standard questionnaire.
- A written account of what the wrong-IPD version felt like. **Most people cannot say what is wrong, only that something is** — which is the lesson.
- An honest statement of what you could not measure and why.

Project 2 — An application people can use (after Module 12)

Interaction, locomotion, and audio, evaluated on real people with real instruments rather than on your own impression.

Requirements

- Hand or controller interaction with grab, release, and at least one tool, with affordances a first-time user discovers unaided.
- At least two locomotion schemes, selectable, with the comfort trade documented.
- Spatialised audio with HRTF, and a demonstration of what is lost without it.
- One VR-specific rendering optimisation applied and measured — foveated rendering, variable rate shading, or stereo instancing.
- A comfort and usability evaluation with at least six participants, including novices.

Done means

- **Six participants' questionnaire scores, reported in full rather than averaged** — including anyone who had to stop.

- A video of a first-time user, with the moments they were confused marked and discussed.
- Before-and-after frame timings for the optimisation, with the visual cost shown honestly.
- **A written account of something that worked for you and not for others.** There will be one, and finding it is the point of testing on other people.

Assessment

Nobody grades this. The standard is therefore mechanical, and you apply it to yourself honestly or the credential is worth nothing.

Component	Weight	Standard
Module exercises	40%	Code runs and produces the specified output. Not 'I understand how I would do it.'
Projects	45%	In the portfolio repository, with a README showing results and a written note on what was hard.
Self-checks	15%	Answered in writing from memory. Recognition is not knowledge; writing exposes the difference.

Required materials

All free. Nothing in this course requires a purchase.

Resource	Role
LaValle — Virtual Reality (free book and Illinois lectures)	The primary text, free and complete. Unusually strong on perception and on the mathematics of tracking, written by someone who worked on the Oculus tracking system.
Stanford EE267 — Virtual Reality, build your own HMD (free)	Course notes, slides, and assignments that build a headset from parts. The optics and display material is the best free treatment of Modules 03 and 04.
Oculus/Meta Developer Documentation — performance and comfort (free)	The practical engineering: frame budgets, timewarp, guardian, and the comfort guidelines. Vendor documentation, and genuinely good on the hard numbers.
Unreal Engine XR documentation (free)	What you will actually build in. Read the performance and stereo-rendering sections before Module 10.
Michael Abrash — blog archive and talks (free)	The latency and display analysis that shaped modern VR. 'Latency — the sine qua non of AR and VR' is required reading for Module 06.
Jerald — The VR Book (chapters and talks free)	The human-factors and design side, which the engineering sources underweight. Good for Modules 07, 08, and 13.

Tooling

- **A headset.** Any PC-connected or standalone consumer headset is sufficient. **You cannot do this course without one** — every claim in it is about an experience.
- **Unreal or Unity** with OpenXR. Do not write a VR runtime; the compositor and timewarp are not useful things to reimplement, and getting them wrong makes people ill.
- **A frame profiler that reports per-eye GPU time** — RenderDoc, PIX, or the engine's own. **Frame time is the subject of this course** and you must be able to see it.
- **A way to record what the user saw**, with timing. Spectator capture plus a log of frame times is enough.
- **Test subjects.** At least six people who are not you, including at least one who has never used VR. **Your own tolerance is not data.**
- **A simulator-sickness questionnaire** (SSQ or the shorter VRSQ). Free, standard, and it makes comfort claims comparable.

Academic honesty, restated for a solo student

There is no one to cheat but yourself, which makes it easier, not harder. Two rules: copy code only after you have written a worse version yourself, and never let a working result stand in for an understood one. When you paste something you do not understand, write a TODO naming what you do not understand, and go back to it.